# Supplementary Material for "Scaled Torus Principal Component Analysis"

**Abstract**

Section A of this Supplementary Material presents a different initialization for $r$. Section B contains further numerical experiments for ST-PCA with regard to the presence of outliers in ST-PCA (Subsection B.1), the applicability in higher-dimensions (Subsection B.2), computational costs (Subsection B.3), further comparisons with T-PCA (Subsection B.4), and limitations of the prediction algorithm (Subsection B.5).

**Keywords:** Dimension reduction; Directional statistics; Multidimensional scaling; Principal component analysis; Statistics on manifolds.

## A A different initialization for $r$

A more sophisticated approach to initialize $r$ is described next. Denote by $\mu$ and $\nu_r$ the two uniform measures on $\mathbb{T}^d$ and $\mathbb{S}^d_r$, respectively. Let $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n \sim \mu$ and $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \sim \nu_r$ be independent and identically distributed samples. Denote by $F_{N,\mathbb{T}^d}$ the empirical cumulative distribution function (ecdf) of the pairwise distances $\{\delta_{\mathbb{T}^d}(\boldsymbol{u}_i, \boldsymbol{u}_j)\}_{1\leq i<j\leq n}$ and by $G_{N,\mathbb{S}^d_r}$ the analogous ecdf for $\{\delta_{\mathbb{S}^d_r}(\boldsymbol{v}_i, \boldsymbol{v}_j)\}_{1\leq i<j\leq n}$, with $N := n(n-1)/2$. Define

$$r^* := \arg\min_{r\geq 0} \mathbb{E}_{\mu\times\nu_r}\left[W_1(F_{N,\mathbb{T}^d}, G_{N,\mathbb{S}^d_r})\right], \quad W_1(F_{N,\mathbb{T}^d}, G_{N,\mathbb{S}^d_r}) = \int_0^1 \left|F^{(-1)}_{N,\mathbb{T}^d}(s) - G^{(-1)}_{N,\mathbb{S}^d_r}(s)\right| \mathrm{d}s, \quad (1)$$

which corresponds to the expected 1-Wasserstein distance between the two pairwise distance ecdfs under the product measure $\mu\times\nu_r$. Above, $F^{(-1)}$ stands for the generalized inverse of $F$. As defined,

*Department of Statistics and Operations Research, University of North Carolina at Chapel Hill (USA).

†Department of Statistics, Carlos III University of Madrid (Spain).

‡Corresponding author. e-mail: pavlos@live.unc.edu.

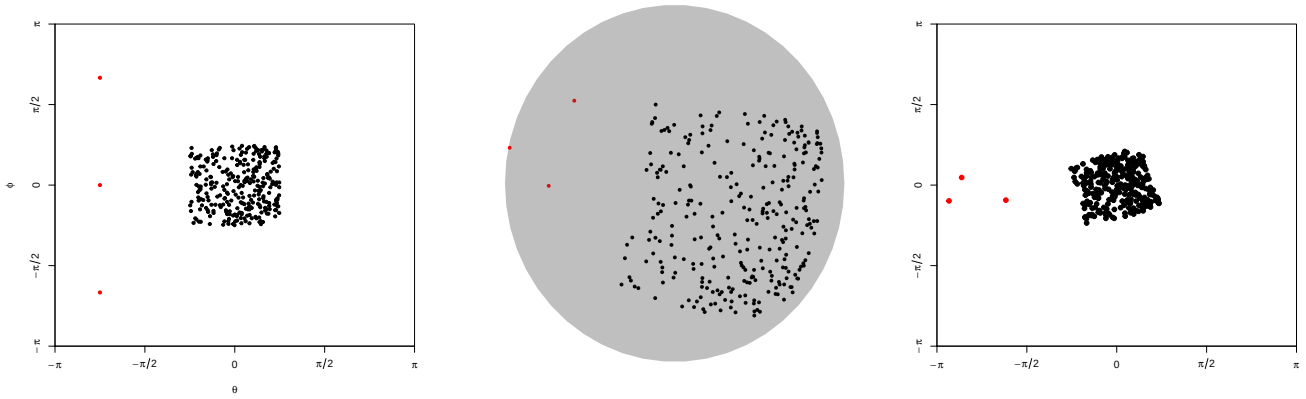
$r^*$ guarantees that, on average, the most spread sample of size $n$ on $\mathbb{T}^d$ can be allocated on $\mathbb{S}^d_r$ while optimally maintaining its interpoint-distance distribution. Uniformly-distributed samples on $\mathbb{T}^d$ can be regarded as the worst-case scenario on which to perform SMDS effectively; $r^*$ is designed to aid precisely in this situation. In practice, the expectation in (1) can be estimated empirically by Monte Carlo and the Wasserstein distance is evaluated by its equivalent form $W_1(F_{N,\mathbb{T}^d}, G_{N,\mathbb{S}^d_r}) = \frac{1}{N}\sum_{i=1}^{N}\left|d_{(i),\mathbb{T}^d} - d_{(i),\mathbb{S}^d_r}\right|$ for $d_{(k),X}$ being the $k$-th order statistic of $\{\delta_X(\boldsymbol{u}_i, \boldsymbol{u}_j)\}_{1\leq i<j\leq n}$. In our experiments, this strategy for initializing the radius to solve (6) was found to yield no remarkable differences when compared to using the faster initialization using $\sqrt{d}$, and thus the latter approach was adopted. Nonetheless, we remark that (1) is significantly more general than the initialization of $\sqrt{d}$, as it can be applied for any transformation from $(X, \delta_X)$ to $(\mathbb{S}^d_r, \delta_{\mathbb{S}^d_r})$.

# B Further numerical experiments

## B.1 Robustness with regard to outliers

Following Figure 5 of Eltzner et al. (2018), we have simulated uniformly on $(-\pi/4, \pi/4)^2 \subset \mathbb{T}^2$ a total of 300 points. In addition, we have added three outliers located at $(-3\pi/4, -2\pi/3)'$, $(-3\pi/4, 0)'$, and $(-3\pi/4, 2\pi/3)'$, for a total of $n = 303$ points. Figure 1a shows the points on $\mathbb{T}^2$.

Recall the trade-off between computational speed and precision for MDS optimization algorithms discussed in Subsection 3.1. In the presence of outliers, good performance follows by using a lower tolerance level (which implies higher precision). For that reason, we use the decreased tolerance level of $10^{-4}$. Figure 1b shows the configuration of points on $\mathbb{S}^2$ obtained by Step 1 of ST-PCA. Finally, Figure 1c shows the scores plot obtained after applying Step 2 of ST-PCA. The outliers can be clearly identified, as well as the shape of the bulk of the points. This performance is comparable to that in Figure 5 of Eltzner et al. (2018).

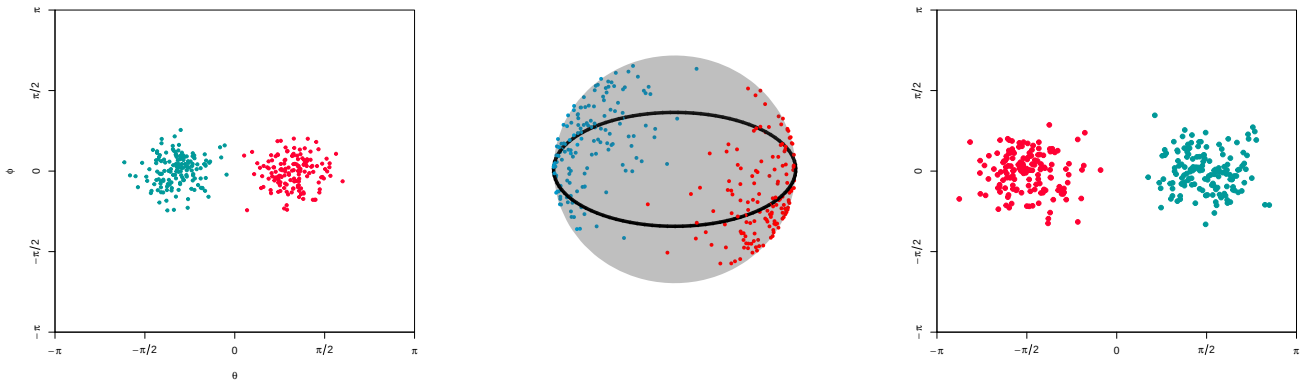



(a) (b) (c)

Figure 1: Panel (a) shows the bulk of points and the outliers on $\mathbb{T}^2$. Panel (b) presents the sample SMDS-mapped to $\mathbb{S}^2$. Panel (c) displays the PNS scores. Shows that the outlier structure is well preserved through ST-PCA

## B.2 Robustness with regard to higher dimensionality

We provide next an example of a high dimensional toy dataset to demonstrate how ST-PCA performs in higher dimensions. For that aim, we simulate data from an isotropic mixture of two wrapped normal distributions in $\mathbb{T}^{50}$. In particular, we simulate $n = 300$ points, where 150 are centered around each of $\boldsymbol{\mu}_1 = (-1, 0, \ldots, 0)'$, $\boldsymbol{\mu}_2 = (1, 0, \ldots, 0)' \in \mathbb{T}^{50}$. Here, visualization of $\mathbb{T}^{50}$ is not feasible and so we show the first two dimensions of $\mathbb{T}^{50}$ in Figure 2a. After running SMDS in Step 1, we have obtained a configuration of points $\widehat{\boldsymbol{y}}_1, \ldots, \widehat{\boldsymbol{y}}_n \in \mathbb{S}^{50}$. For visualization purposes, we show in Figure 2b the spherical view of the configuration's projections to $S^2 \cong \mathbb{S}^2$. For this dataset we have obtained $\widehat{r} = 2.18$. Figure 2c shows the scores plot after applying PNS to $\widehat{\boldsymbol{y}}_1, \ldots, \widehat{\boldsymbol{y}}_n \in \mathbb{S}^{50}$ (Step 2). We remark that due to the high dimensionality of the dataset, the inversion procedure required for Step 3 of ST-PCA and the calculation of the percentage of variance explained by each component is computationally challenging. Nonetheless, the entire cluster structure is effectively captured by the first PNS on $\mathbb{S}^{50}$.



(a) (b) (c)

Figure 2: Panel (a) shows a 2D projection of two clusters in $\mathbb{T}^{50}$. Panel (b) shows the $S^2$ projection of the sample SMDS-mapped to $\mathbb{S}^{50}$. Panel (c) displays the first PNS scores. ST-PCA successfully identifies the two clusters in the first mode of variation.

## B.3 Computational cost with regard to increasing sample size

We explore as follows the computational cost of ST-PCA and its variance explained by the first component on the data generating process from Subsection 4.3. For each of the varying sample sizes, ten samples were generated and the results were averaged. Step 3 of the algorithm was not calculated but is generally computationally cheaper than the calculation of the proportion of variance explained for large datasets. Table 1 summarizes the obtained results. Timings were measured on a (single) 2.40 GHz processor using an implementation of ST-PCA in R.

| $n$ | Step 1 | Step 2 | Comp. var. exp. | Var. exp. |
|---|---|---|---|---|
| 32 | 0.04s | 0.10s | 0.23s | 91.80 % |
| 64 | 0.09s | 0.11s | 0.88s | 88.70 % |
| 128 | 0.31s | 0.14s | 5.00s | 89.30 % |
| 256 | 2.87s | 0.28s | 55.70s | 89.20 % |
| 512 | 30.08s | 0.85s | 9.56min | 88.74 % |
| 1024 | 4.26min | 2.38s | 104.84min | 89.61 % |

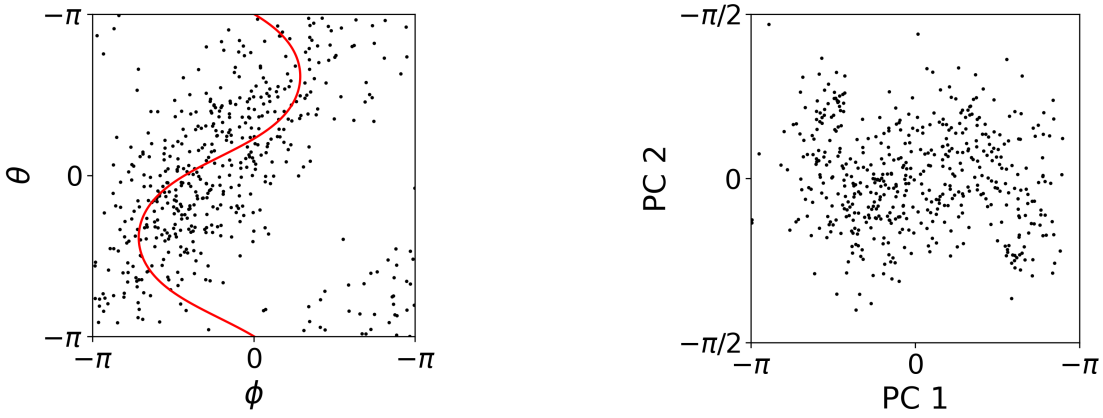
Table 1: Average timings and average percentage of variance explained for the anisotropic wrapped normal generating process of Subsection 4.3. First column: sample size; second and third: running times for Steps 1 and 2 of ST-PCA; fourth: running time for the computation of the percentage of variance explained by the first component; fifth: percentage of variance explained by the first component.

We remark that, as well-known in the literature, the metric MDS operation is at least quadratic

with regard to the sample size. Hence, non-quadratic torus dimensionality reduction techniques might be more suited in practice for very large datasets on $\mathbb{T}^d$.

## B.4 A further comparison with T-PCA

In this subsection we compare the performance of ST-PCA against T-PCA on the same data set as that of Subsection 4.3. Recall that ST-PCA explains 89% of the variance on the first component. The respective statistics for T-PCA are 86% (SO ordering) and 89% (SI ordering), which show that the performance of ST-PCA is comparable to that of the better variant of T-PCA for this dataset and criterion. Figure 3a displays the same dataset as in Figure 4a, with the T-PCA (SI) principal component in red. Figure 3b shows the scores plots for T-PCA (SI); the T-PCA principal component under SO ordering is an analogous sinusoidal curve and so we have omitted it. Note that the ST-PCA principal component (in Figure 4a) is statistically more interpretable and corresponds to what is expected from the data generation process.



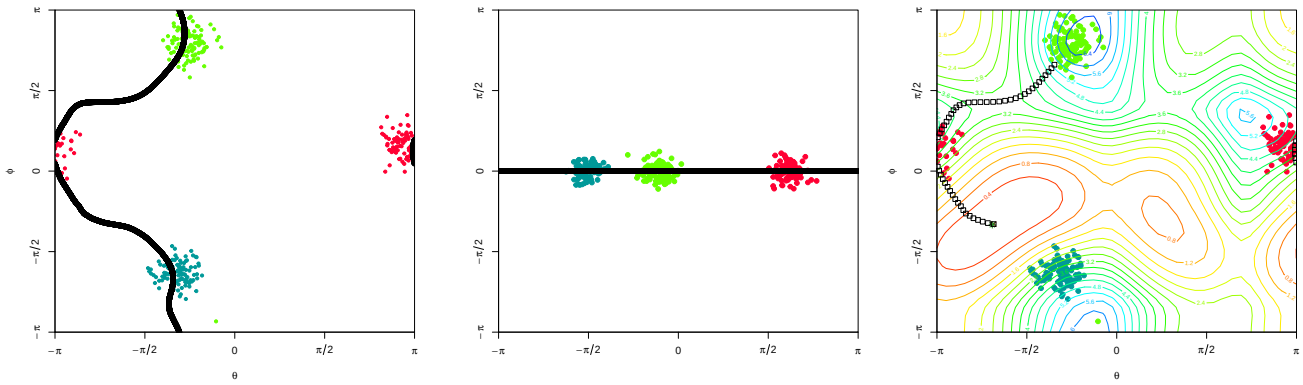

(a) (b)

Figure 3: Panel (a) shows the original data with the T-PCA (SI) principal curve on $\mathbb{T}^2$ in red. Panel (b) displays the first PNS scores for T-PCA (SI). Shows that the T-PCA principal component is less intuitive than the ST-PCA principal component.

## B.5 Limitations of the prediction algorithm and importance of the initialization

The prediction algorithm described in Subsection 3.4 consists on solving a sequence of optimization problems and interpolating those solutions to obtain a curve. Although the resulting collection of points has the advantage of being very flexible (see, e.g., Subsection 5.1), it can also yield discontinuities due to the existence of multiple minima (global or local) of the objective function in (8).

We now describe an example where this optimization problem can be challenging by showing the existence of multiple minima through a contour analysis of (8). We generate another instance of the data in Subsection 4.1 and perform the first step of ST-PCA with a lower relative tolerance level of $10^{-4}$ for increased accuracy, followed by Steps 1–2 of ST-PCA. Figure 4a displays the three original clusters in $\mathbb{T}^2$ in red, green, and blue, as well as the principal curve in black. Figure 4b shows the scores plot on $\mathbb{S}^2$ after Steps 1–2. We then perform Step 3 for an equispaced grid $\{\boldsymbol{\xi}_j\}_{j=1}^{100} \subset S^1$. Figure 4c shows a contour analysis for the objective function in (8) during the prediction of $\boldsymbol{\xi}_{56}$ of the principal curve, prior to interpolation. Here, the prediction of $\boldsymbol{\xi}_1$, $\widetilde{T}(\boldsymbol{\xi}_1)$, corresponds to the northernmost black point of the plot and the path is moving south in a continuous manner. Warmer colors of the contours correspond to lower values of the objective function (8). Note that an unsensible initialization of the initial value for this optimization problem could easily result in a solution that is different from the global minimum (e.g., the local minimum approximately at $(\pi/4, -\pi/4)'$). This would result in a discontinuity in the principal component between $\widetilde{T}(\boldsymbol{\xi}_{55})$ and $\widetilde{T}(\boldsymbol{\xi}_{56})$. In the example, the continuity on the resulting principal component is induced due to the initialization of (8) for $\boldsymbol{\xi}_{56}$ at the previously-obtained solution, $\widetilde{T}(\boldsymbol{\xi}_{55})$ (in the southeastern region of Figure 4c).




(a) (b) (c)

Figure 4: Panel (a) shows the original data with the principal curve on $\mathbb{T}^2$. Panel (b) displays the first PNS scores. Panel (c) presents a contour analysis for the prediction of the 56th point of the principal component on the torus, where "x" (in red) denotes the solution to the optimization problem, and "*" (in dark green) the minimum value obtained by the contours. Shows the existence of multiple local minima in the inversion process of the principal curve.

# References

Eltzner, B., Huckemann, S., and Mardia, K. V. (2018). Torus principal component analysis with applications to RNA structure. *Ann. Appl. Stat.*, 12(2):1332–1359.